FIN2020 HW3 文杰 123090612

Terminal wealth is (stocks + bond):

$$\tilde{Y}_1 = (1+r_f)(Y_0 - a) + a(1+\tilde{r}) = Y_0(1+r_f) + a(\tilde{r} - r_f),$$

so the optimal a^* solves the vN–M expected-utility problem $\max_a E[u(\tilde{Y}_1)]$ with FOC

$$E\{u'[Y_0(1+r_f) + a^*(\tilde{r} - r_f)](\tilde{r} - r_f)\} = 0.$$

Here: $Y_0 = 100$, $r_f = 0.10$, $r_G = 0.30$, $r_B = 0.05$, and $\Pr(G) = \Pr(B) = 1/2$. Define

$$Y_G(a) = 110 + 0.20a, \qquad Y_B(a) = 110 - 0.05a.$$

Question 1: $u(Y) = \ln Y$

Maximize

$$\frac{1}{2}\ln(110 + 0.20a) + \frac{1}{2}\ln(110 - 0.05a).$$

FOC:

$$\frac{1}{2}\frac{0.20}{110 + 0.20a} + \frac{1}{2}\frac{-0.05}{110 - 0.05a} = 0$$

$$\Longrightarrow \frac{0.20}{110 + 0.20a} = \frac{0.05}{110 - 0.05a}.$$

$$\therefore a^* = 825$$

Therefore, the log-utility investor allocates

$$\boxed{a_1^* = 825.}$$

a may exceed Y_0 if borrowing/margin is allowed.

Question 2: $u(Y) = \dfrac{Y^{1-\gamma} - 1}{1-\gamma}$ with $\gamma = 2$

With $\gamma = 2$,

$$u(Y) = \frac{Y^{-1} - 1}{-1} = 1 - \frac{1}{Y}.$$

Expected utility (up to an irrelevant constant) is

$$E[u(\tilde{Y}_1)] = 1 - \frac{1}{2}\left(\frac{1}{Y_G(a)} + \frac{1}{Y_B(a)}\right).$$

FOC:

$$\frac{d}{da}\left[-\frac{1}{2}\left(\frac{1}{Y_G(a)} + \frac{1}{Y_B(a)}\right)\right] = 0$$

$$-\frac{1}{2}\left(-\frac{0.20}{Y_G(a)^2} + \frac{0.05}{Y_B(a)^2}\right) = 0 \Longrightarrow \frac{0.05}{Y_B(a)^2} = \frac{0.20}{Y_G(a)^2}.$$

Thus $Y_G(a)^2 = 4Y_B(a)^2$, and since wealth is positive, $Y_G(a) = 2Y_B(a)$:

$$110 + 0.20a = 2(110 - 0.05a) = 220 - 0.10a$$

$$\Rightarrow a^* = \frac{1100}{3} \approx 366.67.$$

Therefore, the $\gamma = 2$ (CRRA) investor allocates

$$\boxed{a_2^* = \frac{1100}{3} \approx 366.67.}$$

Which investor is more risk averse? Who invests more in stocks?

For CRRA utility $u(Y) = \frac{Y^{1-\gamma}-1}{1-\gamma}$, the coefficient of **relative** risk aversion is constant and equals γ, and $\gamma = 1$ corresponds to log utility.

- **Investor 1 (log)**: $\gamma = 1 \Rightarrow R_R = 1$.
- **Investor 2 ($\gamma = 2$)**: $R_R = 2$.

Therefore, **Investor 2 is more risk averse** (higher relative risk aversion).

Arrow's comparative risk aversion result implies higher risk aversion leads to a smaller optimal risky allocation.

Numerically:

- Investor 1: $a_1^* = 825$.
- Investor 2: $a_2^* \approx 366.67$.

Investor 1 allocates more to stocks (more leverage / larger risky position), and **Investor 2 allocates less**.